Abstract Algebra | Quiz 8 | October 28, 2026

Name: _____ | Hammack | Score: _____

1. Suppose $\varphi : \mathbb{Z} \to \mathbb{Z}_5 \times \mathbb{Z}_6$ is a homomorphism, and $\varphi(1) = (3, 2)$.

   (a) $\varphi(3) =$

   (b) $\ker(\varphi) =$

2. Recall that the determinant function $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$ is a surjective homomorphism.

   (a) $\ker(\varphi) =$

   (b) According to the First Isomorphism Theorem, $GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong$

3. Prove that a homomorphism $\varphi : G \to H$ is injective if and only if $\ker(\varphi) = \{e\}$.